

EDUCATION

Harvey Mudd College, Claremont, California (GPA: 3.36/4.00)
Bachelor of Science in Engineering with Distinction

May 2026

RELEVANT COURSEWORK

Mechanical Design, Materials Engr., Manufacturing Planning/Execution, Mechanics & Wave Motion, Multivariable Calculus, Data / Probability & Stats for Engr., Differential Equations, Fluid Mechanics, Electrical & Magnetic Circuits / Devices, Thermodynamics, Intro to Engr. Design & Manufacturing, Advanced Systems Engr., Continuum Mechanics, Computer Science

SKILLS

CAD/CAE Software: SolidWorks, AutoCAD, Fusion, Onshape, COMSOL Multiphysics

Programming & Data Analysis: MATLAB, Python, LabView, SystemVerilog, Javascript, C, Arduino (C++), SSH

Advanced 3D Printing (SLA, SLS), CNC Machining (ShopBot, Mill, Lathe), Laser Cutting, Welding, Soldering, and comprehensive manual shop equipment (Saws, Grinders)

Languages: Spanish (Native) and French (Limited Proficiency)

EXPERIENCE

Blue Origin, Claremont, California

August 2025 - May 2026

Mechanical Team Lead - Automated ground-side umbilical mating procedure for New Glenn rocket

- Led mechanical sub-team in engineering an actuation system to autonomously mate large umbilical panels, addressing critical mission requirements for millimeter alignment accuracy and efficiency.
- Drove mechanical design, conducted mechanism trade studies, and performed CAD structural analysis (FEA).
- Coordinated with multiple engineering disciplines (controls, sensors) to define system architecture, plan hardware installation, and execute systems integration with the master control algorithm.

Flow Imaging Lab Research, Claremont, California

August 2024 - May 2026

CAD Lead & Trajectory Designer - Research on archerfish-inspired propulsion system for aerial-aquatic robots

- Designed and fabricated improvement to existing experimental apparatus and mounting fixtures using SolidWorks, serving as lab's CAD Manager.
- Quantified hydrodynamic forces and torques during water-exit events utilizing a 6-axis load cell.
- Analyze fluid dynamics in Matlab by implementing Particle Image Velocimetry to visualize flow patterns and utilizing Computational Fluid Dynamics to validate experimental data and inform trajectory design.
- Maintain and optimize data acquisition equipment and experimental tank to assure operational readiness and data integrity.

Harvey Mudd Automotive Club (MACH), Claremont California

September 2022 - May 2026

Founding Member & Mechanical Lead, Custom Automotive Build

- Directed the mechanical integration of a custom vehicle build, mounting an internal combustion engine and coupling the drivetrain to the rear axle.
- Designed and installed complete steering and braking subsystems, establishing the physical and structural baseline for a planned hybrid-electric powertrain implementation.

Mudd Rocketry Club, Claremont, California

September 2022 - May 2026

Mechanical Engineer - Iterating on previous rocket models by analyzing failure modes

- Launched and recovered Gladius-III reaching 17172 feet with 99.95% altitude accuracy.
- Fabricated and installed composite flight hardware to tight tolerances applying GD&T principles across milling, waterjet, and laser cutting operations.
- Evaluated interface compatibility and executed integration and test planning alongside Avionics and Recovery sub-teams to ensure seamless system assembly.

Indomo TX, Claremont California

January 2025 - May 2025

Mechanical Engineer - Redesigned a next-generation autoinjector for a clinical-stage company's novel, at-home acne treatment

- Engineered mechanical improvements, defined system assembly plans, and optimized the device for Design for Manufacturability.
- Implemented a critical safety interlock integrating user activation mechanism with a skin-contact sensor, eliminating risk of accidental device discharge.
- Developed a decoupled mechanism to separate needle deployment from drug delivery, calibrating system's mechanical timing to enhance user safety and ensure reliable operation.
- Validated subsystem CAD designs through rapid 3D prototyping to accelerate development cycles.